3 W Power LED

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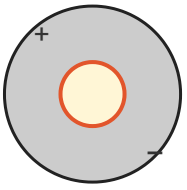
Cool white · on 20 mm aluminium star (generic type)

Datasheet summary compiled from typical manufacturer data (T_A = 25 °C unless noted) – for teaching purposes. Not an official manufacturer datasheet; values of real parts may differ slightly.

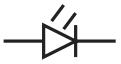
Features

- High-power LED, cool white (approx. 6500 K)
- Needs a heat sink at full current
- Constant-current drive recommended
- **Not designed for reverse operation**

Package



20 mm aluminium star (MCPCB)
solder pads marked + and –
mount on a heat sink



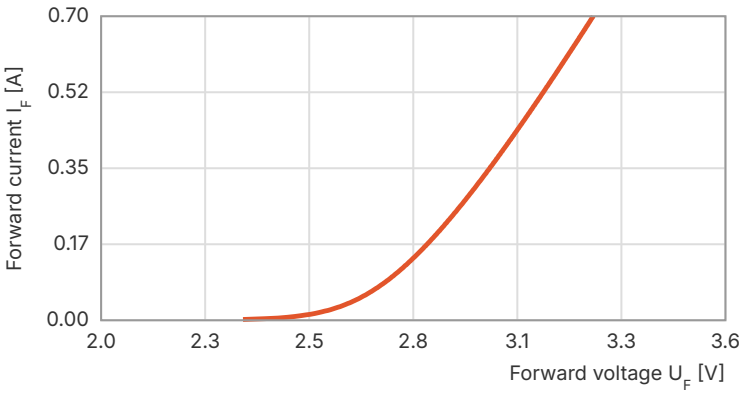
Absolute maximum ratings

Parameter	Symbol	Value	Unit
Forward current (DC)	I_F	700	mA
Pulse forward current (t ≤ 0.1 ms, duty 1/10)	I_{FP}	1000	mA
Reverse voltage	U_R	5	V
Power dissipation	P_D	3.2	W
Junction temperature	T_j	125	°C

Electrical characteristics

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Unit
Forward voltage	U_F	I_F = 350 mA	2.8	3.0	3.4	V
Forward voltage	U_F	I_F = 700 mA	2.9	3.2	3.6	V
Reverse current	I_R	U_R = 5 V	–	–	10	µA
Luminous flux	Φ_V	I_F = 700 mA	180	200	–	lm
Viewing angle (2θ½)	–	I_F = 350 mA	–	120	–	°

Typical forward characteristic (T_j = 25 °C)



Typical curve at 25 °C. The forward voltage rises with the current; at higher junction temperature it becomes slightly lower.